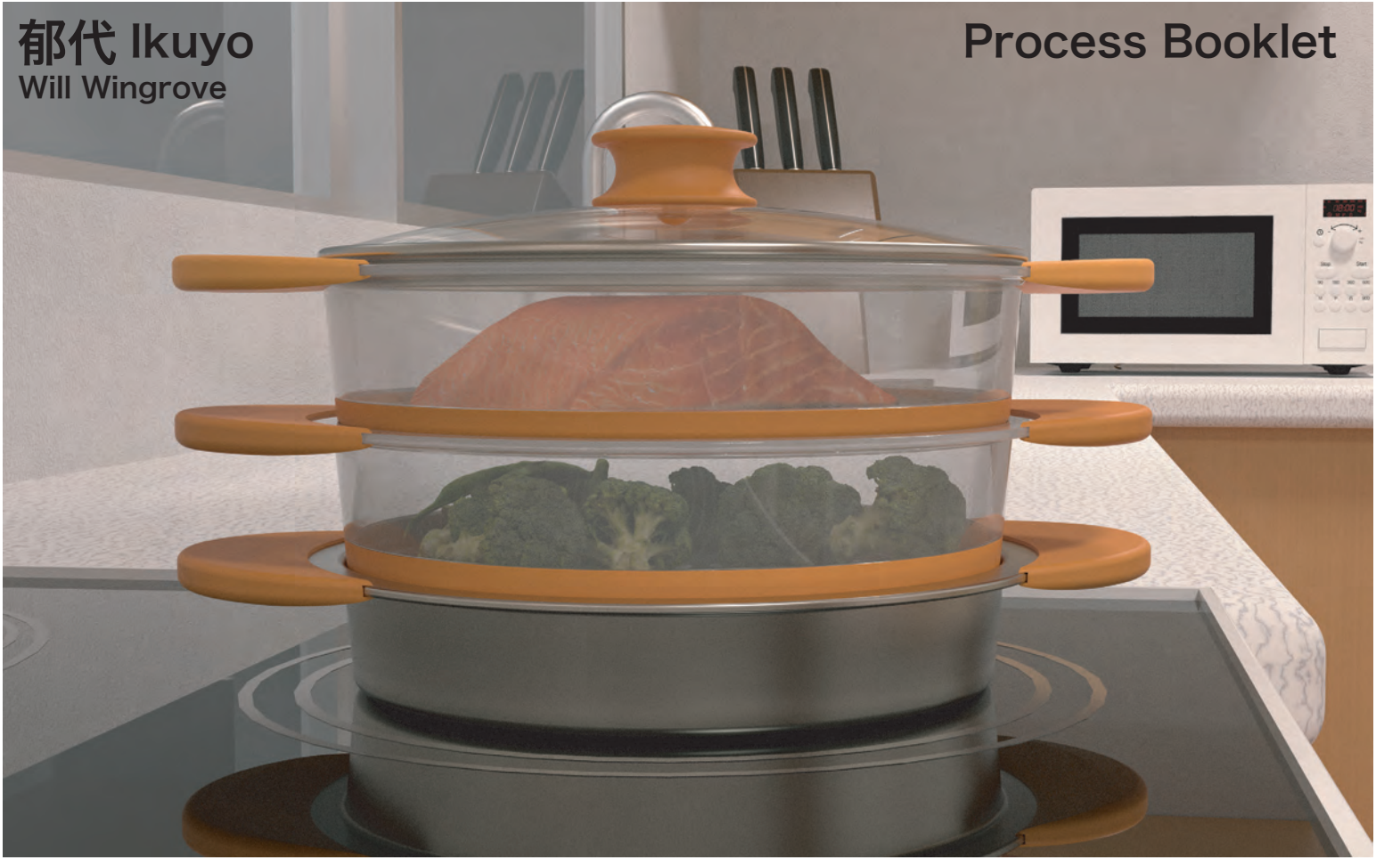
郁代 Ikuyo

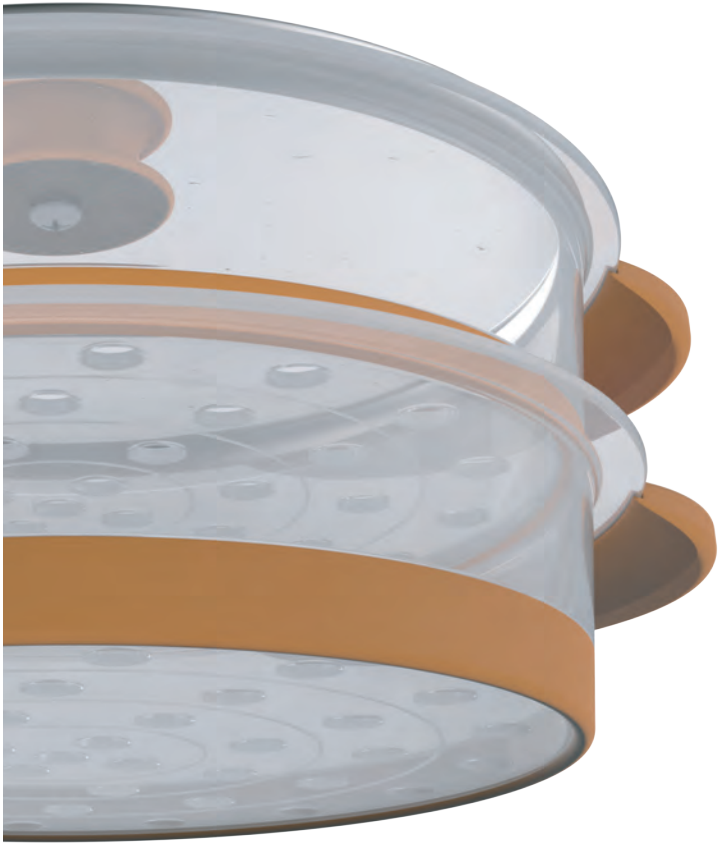
Will Wingrove

Process Booklet



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Page 2 User & Environment
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Page 8 Ideation Phase
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Project Introduction

Steaming is one of the most accessible and reliable cooking methods, yet many novice cooks, especially students and young adults living in small spaces, struggle to cook with confidence. Existing products often feel intimidating, require multiple parts, or create safety concerns related to steam, heat, and handling. In compact, shared kitchens, even simple setup and cleanup steps can become overwhelming.

This project explores how a stovetop steamer can be redesigned to better support beginner cooks by simplifying the process, reducing risk, and improving usability in tight living environments. The goal is to develop a compact, intuitive, and reassuring steaming solution that makes healthy meal preparation easy and approachable for new users.

User & Enviornment

The primary users are novice cooks with limited experience, including students and young adults transitioning to independent living. Many rely on takeout or pre-made meals and often feel uncertain or intimidated when preparing food from scratch. They value convenience, reassurance, and simple workflows over precision or advanced control.

These users typically cook in small, shared apartment or dorm-style kitchens where storage is limited, counter space is scarce, and stovetop access is often shared or restricted. Tight kitchen layouts make multitasking more stressful and increase the risk associated with handling hot cookware and escaping steam.



*Generative AI

What is Ikuyo?

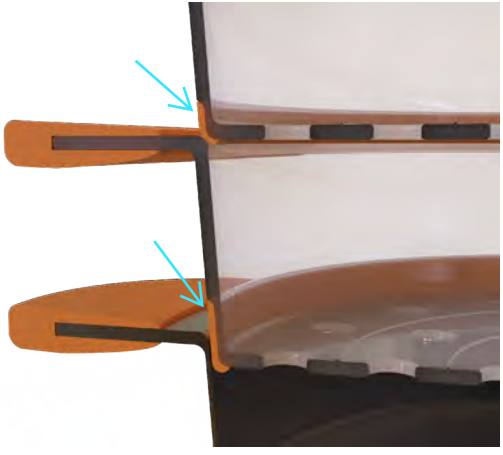


Ikuyo addresses the main challenges faced by new cooks by combining safe handling, efficient meal capacity, and compact storage into one approachable design. Its multi-tiered baskets allow users to cook an entire meal at once, while the heat-safe silicone grips reduce the risk of burns. The collapsed form fits easily into crowded drawers, and the smooth interior surfaces make cleanup fast and stress-free, removing the intimidation, clutter, and safety concerns that often discourage beginners from cooking.









Co-moulded gaskets keeps steam in, regulates teperatures, and ensures safe handling in motion.

Soft touch silicone grips keeps the heat away from your hands

With a combined internal volume of 10 cups across both baskets, the steamer provides enough capacity for two people to comfortably cook an entire meal.

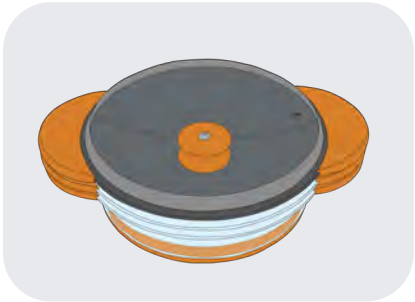
Context Renders

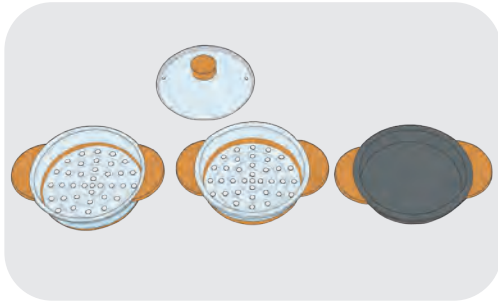




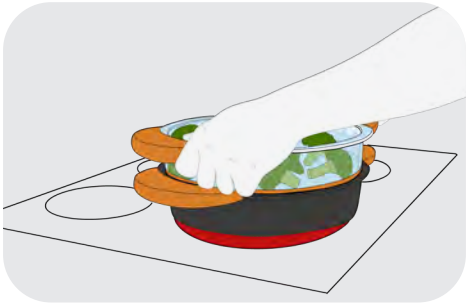


Use Cycle

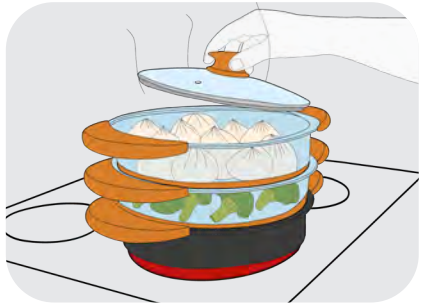


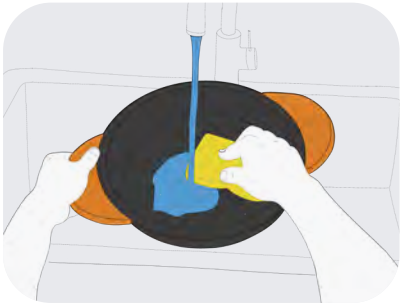


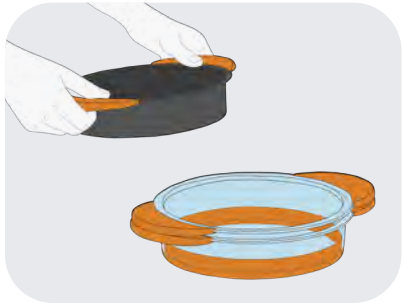


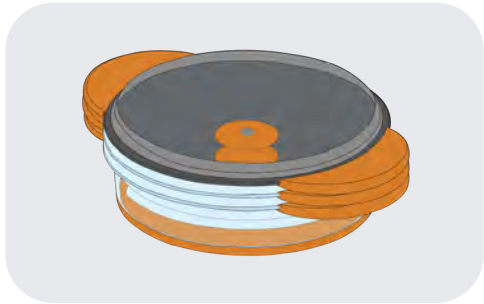


















Reviewed common stovetop, electric, and bamboo steamers in stores/online, focusing on materials, stacking systems, handles, and lids.

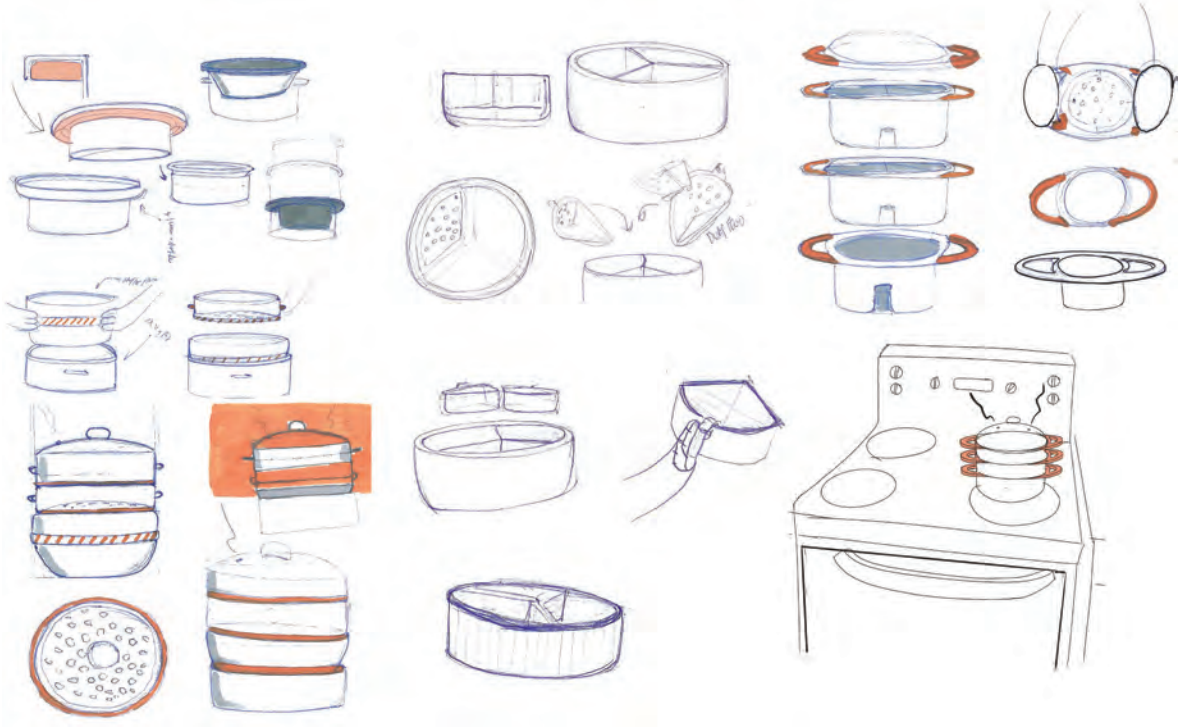
Identified key issues for novices: hot/awkward handles, steam leaks, confusing setup, and bulky products that are hard to store in small shared kitchens.





Existing Product Comparison

Ideation Phase



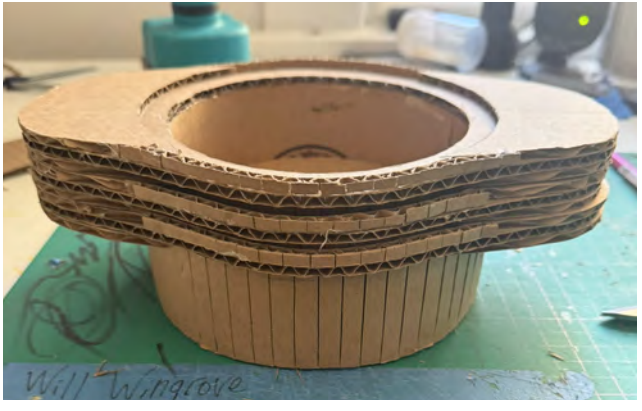
These sketches represent three early stovetop steaming concepts, each exploring different forms, mechanisms, and usability approaches. Each direction offered unique advantages and limitations, which were evaluated through quick prototypes. These early studies helped clarify which features best supported novice cooks and informed the next stage of development.

Prototyping .I









The first round of prototypes was developed around the three-stack standalone concept. The goal was to explore a compact pot system sized for a single user, capable of cooking a complete meal. To refine the proportions and overall footprint, a large number of scale and form models were created.

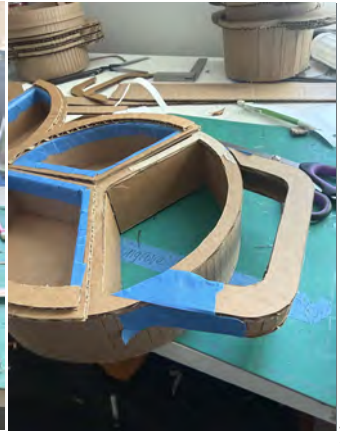
Prototyping .II







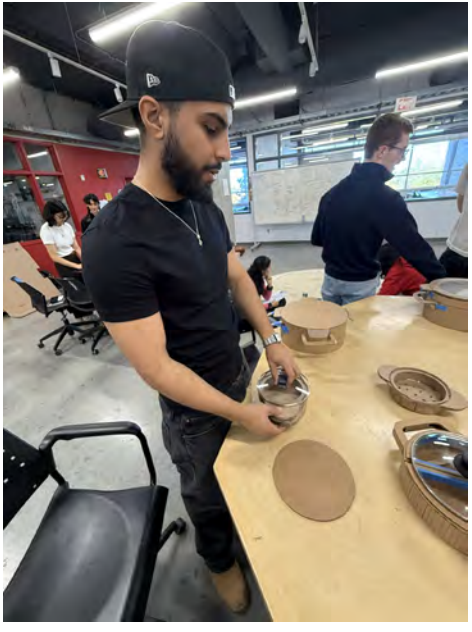




The second round of prototypes explored a single-layer, multi-chamber design. This version featured a trisected interior that allowed users to steam different food groups separately, preventing cross-contamination while keeping flavors distinct. The low-profile form was intentionally kept roughly the same height as the collapsed first-round prototype, making it compact, easy to store, and suitable for small kitchens.

Testing Prototype I

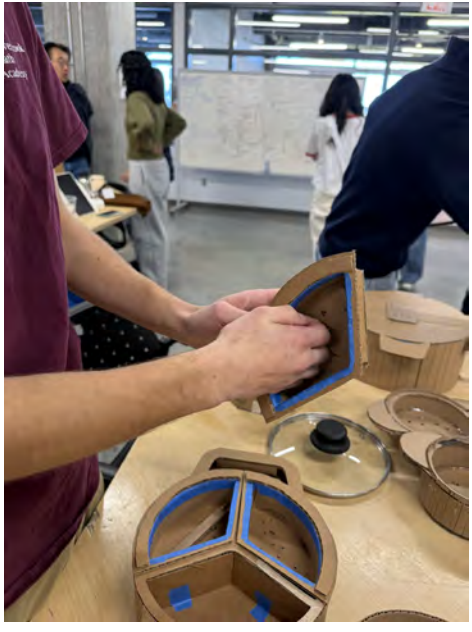


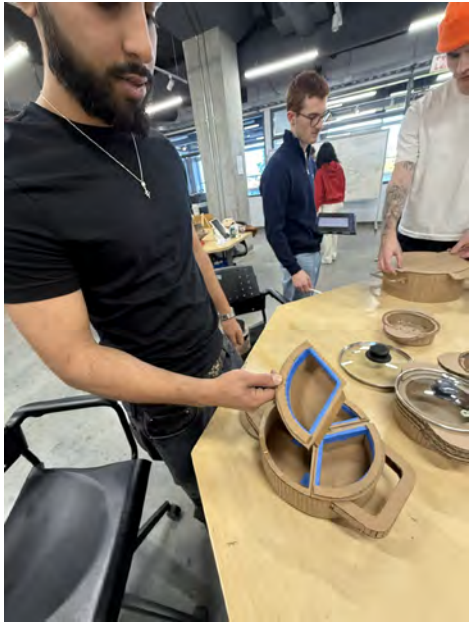




For the first (smaller) prototype, most participants really liked the handle design. The grip size felt secure and comfortable, and people reported feeling confident about how they would lift and move the pot. However, many participants felt that the overall capacity was too small, especially for cooking a complete meal for two people. A frequent comment was that it seemed ideal for a single person or side dishes, but not for a shared main meal. The most common recommendation was to increase the diameter of the pot while keeping the same handle design and depth, effectively scaling up the footprint without changing the way it feels to hold.

Testing Prototype II



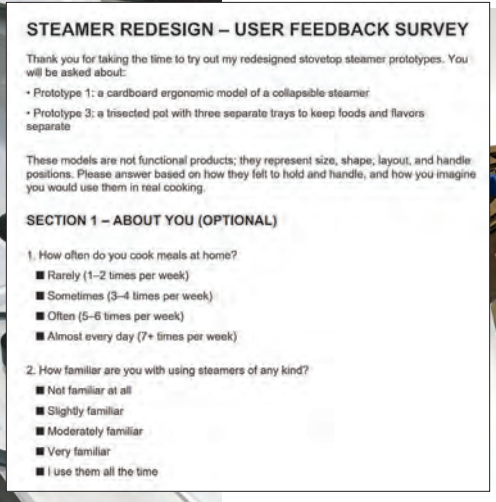




For the trisected / multi-compartment prototype, participants responded positively to the overall concept of separating foods and preventing flavor contamination. Many could imagine use cases like keeping proteins and vegetables apart, or accommodating different preferences at the same time. However, they raised more concerns about the size and proportions of this version. The handles were often described as comfortable but unnecessarily large, which made the product feel bulkier than it needed to be. Several participants questioned whether the pot's proportions and footprint were appropriate for their kitchens and everyday cooking routines.

When asked to compare, most participants preferred Prototype 1 overall, seeing it as more immediately practical and easier to imagine in their everyday cooking. Rather than abandoning it, they suggested enlarging the diameter so it could better support cooking for two while retaining the successful handle ergonomics. Prototype 2 was developed in response to this feedback, exploring a larger overall size and profile to provide greater capacity. The final steamer concept emerged through a convergent process that combined the strongest elements of both prototypes: the overall size and profile were derived from Prototype 2 to achieve a larger capacity, while the three-stacking system and mechanism evolved from Prototype 1, and its well-liked handles were adapted to fit the larger form factor. Together, these convergent decisions resulted in a compact, large-capacity steamer that balances kitchen footprint, meal volume, and ergonomic usability.





STEAMER REDESIGN – USER FEEDBACK SURVEY

Thank you for taking the time to try out my redesigned stovetop steamer prototypes. You will be asked about:

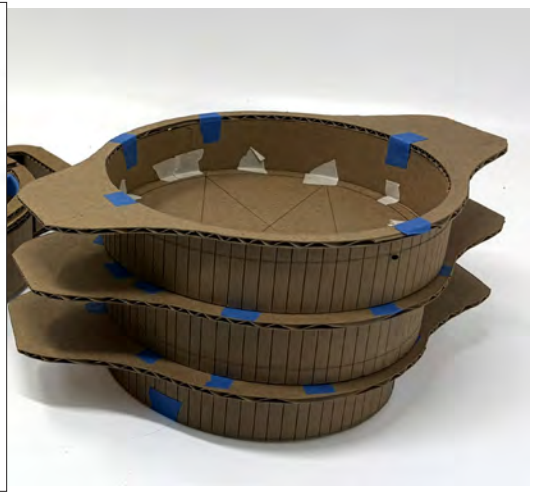
- Prototype 1: a cardboard ergonomic model of a collapsible steamer
- Prototype 3: a trisected pot with three separate trays to keep foods and flavors separate

These models are not functional products; they represent size, shape, layout, and handle positions. Please answer based on how they felt to hold and handle, and how you imagine you would use them in real cooking.

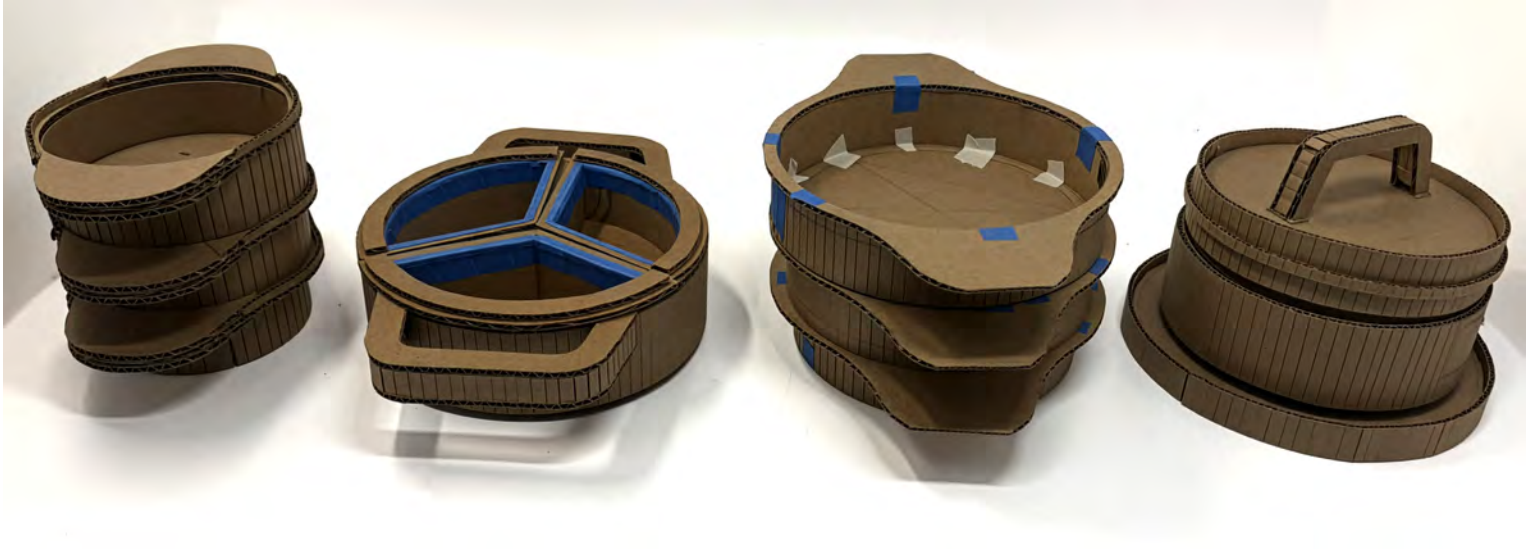
SECTION 1 – ABOUT YOU (OPTIONAL)

1. How often do you cook meals at home?
   - ■ Rarely (1–2 times per week)
   - ■ Sometimes (3–4 times per week)
   - ■ Often (5–6 times per week)
   - ■ Almost every day (7+ times per week)

2. How familiar are you with using steamers of any kind?
   - ■ Not familiar at all
   - ■ Slightly familiar
   - ■ Moderately familiar
   - ■ Very familiar
   - ■ I use them all the time



Summary of User Feedback



Early prototypes were constructed from cardboard based on the initial sketches to explore variations in size, shape, handles, and lid configurations. These quick models allowed for fast, low-risk testing of ergonomics and usability, helping identify which directions best supported the design critera.



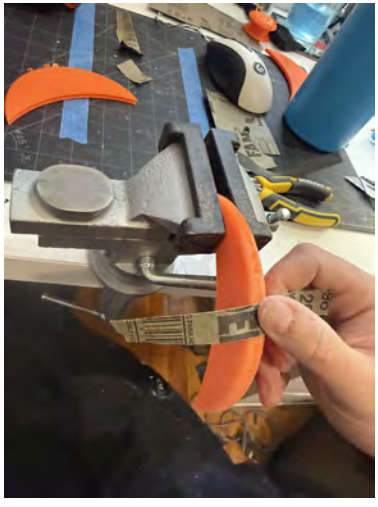



Two bucks were turned on the lathe, vacuum formed, then trimmed on the bandsaw.

Top sections were laser cut and sanded; the bottom gasket detail was created by masking with electrical tape and painting.

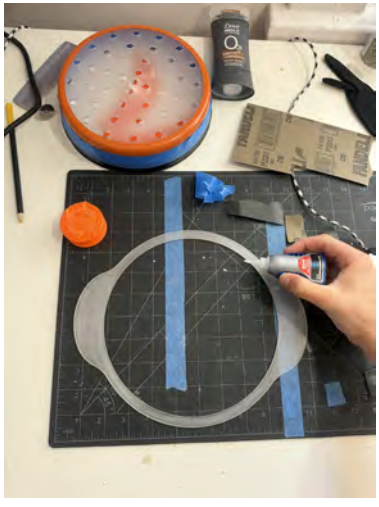
Top and base pieces were then assembled and glued together using super glue.













The base section was built up with Bondo, sanded smooth, and painted to resemble stainless steel.

Handles were 3D printed, filled in with Bondo, sanded, and painted orange to resemble soft touch silicone rubber.

Final Build Process



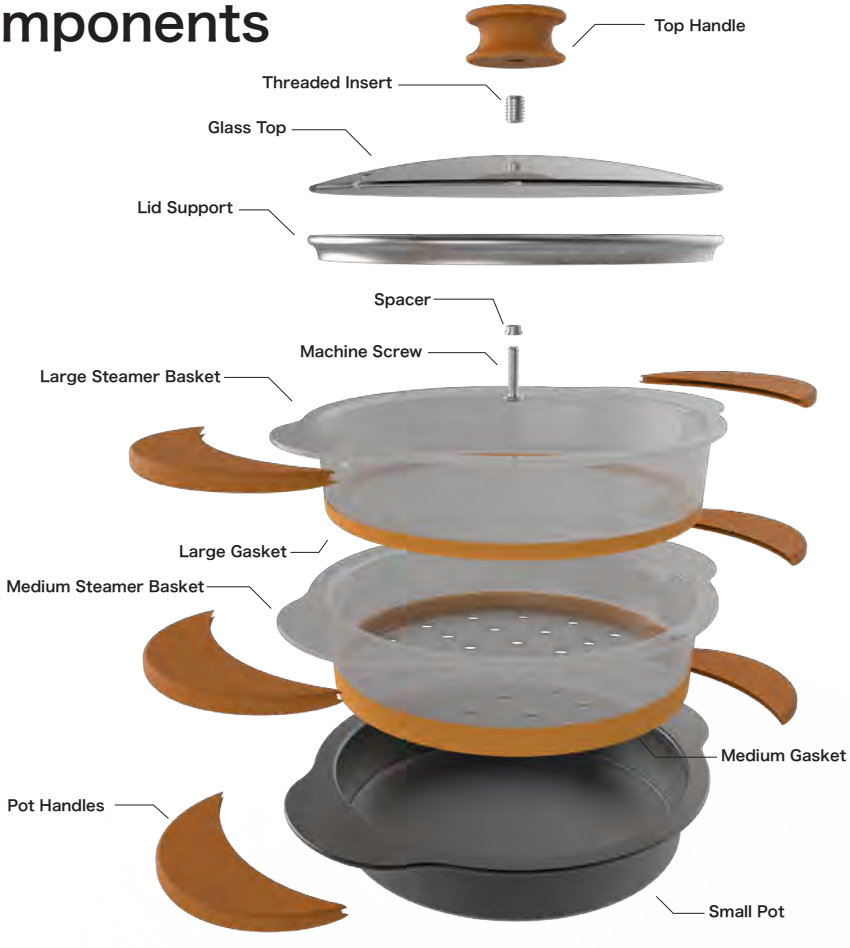


Finished Model



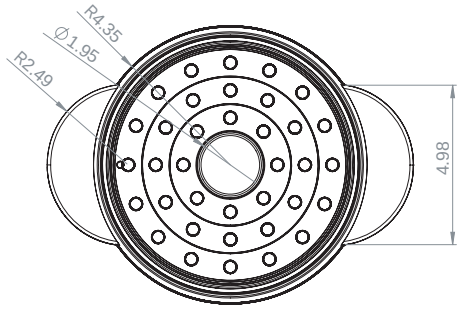


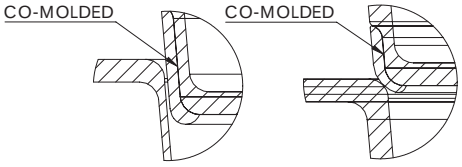
Components



Detailed Parts Summary

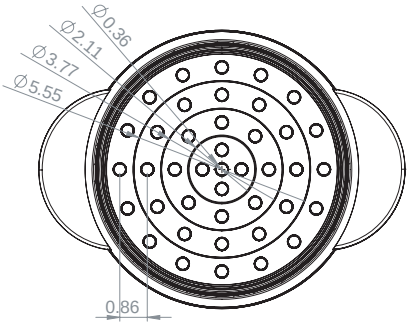
Name	Material	Notes / Purpose
Top Handle	Food-grade silicone rubber	Heat-safe, soft, non-slip grip
Threaded Insert	304 stainless steel	Strong, corrosion-resistant threads, 18-8
Glass Top	Tempered borosilicate glass	Durable, heat-resistant lid material
Metal Ring (Lid Support)	Stainless steel	Reinforces and secures the glass lid
Spacer	Nylon (PA66)	Separates hardware, prevents wear
Phillips Screw	304 stainless steel	Machine screw 18-8
Large Steamer Basket	PCTG co-polyester	Clear, durable, BPA-free, over-moldable
Large Gasket	Food-grade silicone rubber	Co-molded seal with large basket
Medium Steamer Basket	PCTG co-polyester	Same material as large basket
Medium Gasket	Food-grade silicone rubber	Forms seal for medium basket
Small Pot	18/0 steel (304)	Standard cookware alloy, durable
Pot Handles	Food-grade silicone rubber	Six total; heat protection & comfort



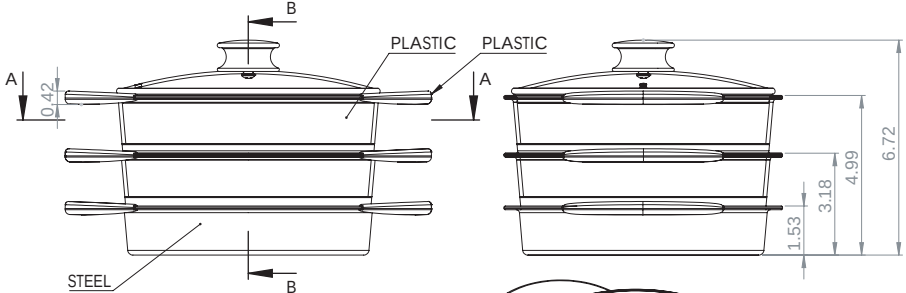


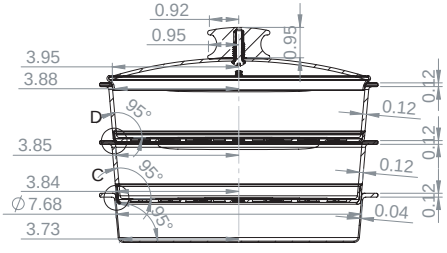
DETAIL C
SCALE 2 : 1

DETAIL D
SCALE 2 : 1

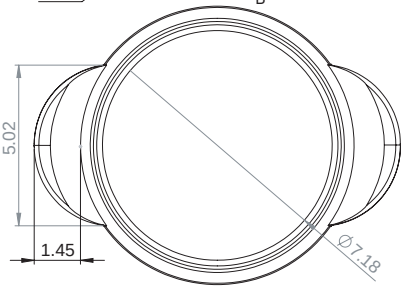


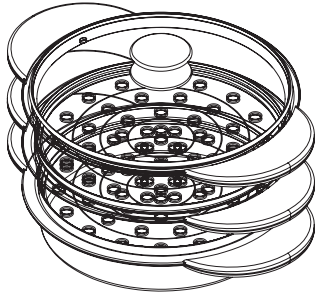
SECTION A-A





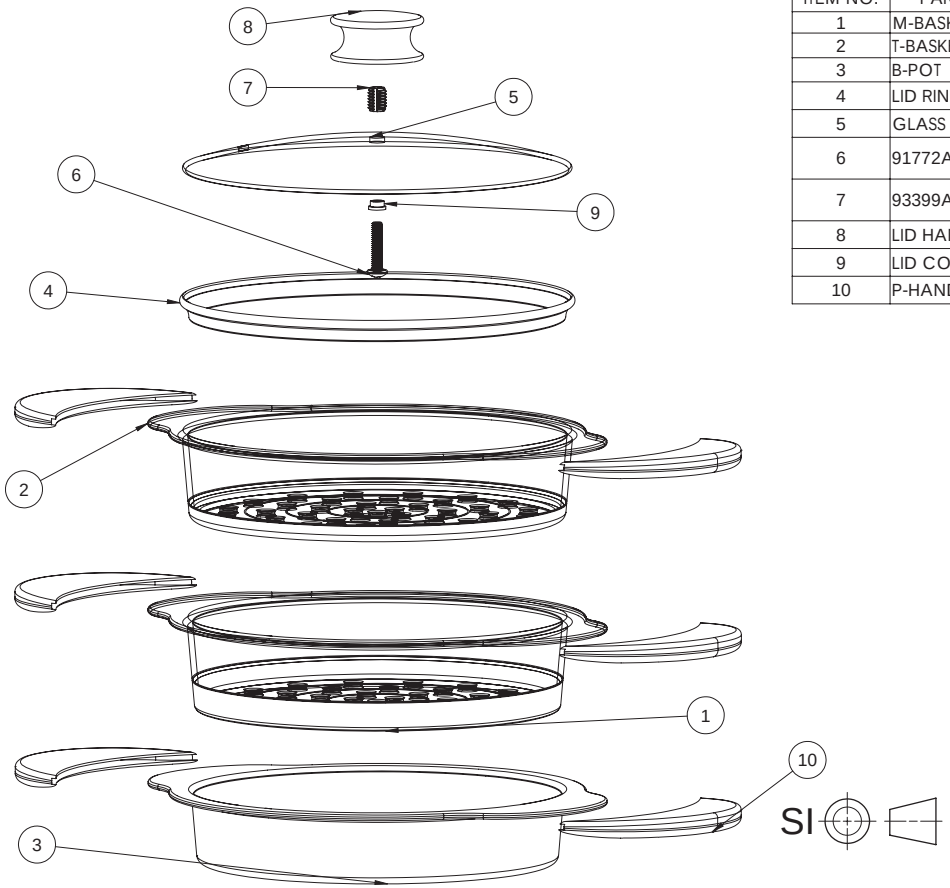
SECTION B-B





SI

CARLETON SCHOOL OF INDUSTRIAL DESIGN OTTAWA, CANADA			
TITLE: IKUYO			TOLERANCES: DECIMAL ± 0.05 ANGLE ± 0.5°
DRAWN: WILL WINGROVE			
CLIENT: TIM HAATS			UNITS: IN
DWG NO: --		REV: 1	SCALE: 1:3
DATE: 11/24/2025		SIZE: 11x17	SHEET 1 OF 2



ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	M-BASKET	PCTG CO-POLYESTER	1
2	T-BASKET	PCTG CO-POLYESTER	1
3	B-POT	18/0 Steel (304)	1
4	LID RING	304 Stainless Steel	1
5	GLASS	Tempered Borosilicate Glass	1
6	91772A296	PASSIVATED 18-8 STAINLESS STEEL PAN HEAD PHILLIPS SCREW	1
7	93399A108	18-8 STAINLESS STEEL TAPPING INSERTS FOR PLASTIC	1
8	LID HANDLE	FOOD GRADE SILICONE	1
9	LID COLLAR	NYLON (PA66)	1
10	P-HANDLE	FOOD GRADE SILICONE	6

SI

CARLETON SCHOOL OF INDUSTRIAL DESIGN
OTTAWA, CANADA

TITLE:	IKUYO		TOLERANCES: DECIMAL ± 0.05 ANGLE ± 0.5°
DRAWN:	WILL WINGROVE		
CLIENT:	TIM HAATS		UNITS: IN
DWG NO: 1		REV: 1	SCALE: 1:3
DATE:	11/24/2025	SIZE: 11x17	SHEET 2 OF 2

